Final Project Report MATH 563

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Abstract

give some background

1 Introduction

1.1 Notation

1.2 Preleminaries

We first note an important relation between the proximal and subdifferential operators.

Proposition 1. Given a function $f \in \Gamma$, we have

$$\mathbf{prox}_{\lambda f} = (I + \lambda \partial f)^{-1}, \tag{1}$$

where $(I + \lambda \partial f)^{-1}$, for $\lambda > 0$, is called the **resolvent** of the operator ∂f. In particular, the proximal operator is the resolvent of the subdifferential operator [3].

Definition 1 (Maximal Monotone operator)**.** A monotone operator is **maximal** if its graph is not properly contained in the graph of any other monotone operator. If A is monotone (resp. maximal monotone) then so are A^{-1} and λA if $\lambda > 0$.

The next proposition, given in [2], will be useful in the construction of the optimizing algorithms.

Proposition 2. If F is maximal monotone, then $(I + \lambda F)^{-1}(\hat{x})$ exists and is unique for all $\hat{x} \in R^n$. The value $x = (I + \lambda F)^{-1}(\hat{x})$ of the resolvent is the unique solution of the monotone inclusion

$$\hat{x} \in x + \lambda F(x)$$

2 Algorithms

The algorithms we present are used to solve the non-blind image deblurring problem, which is a convex optimization problem of the form

$$\begin{aligned} &\min_{x,y} f(x) + g(y) \\ &\text{subject to } Ax = y, \end{aligned} \tag{2}$$

where $f, g \in \Gamma$ have inexpensive prox-operators. We are given a vectorized, blurred image b, along with the kernel K used to blur it using the model

$$Kx + \eta = b, \tag{3}$$

Preprint. Under review.

where x is the original image, and η the added noise. Our goal is to recover the original image x. We formulate the problem in two ways, namely, one which uses the L^1 norm, and the other the L^2 norm. First,

$$\min_x \|Kx-b\|_1 + \delta_S(x) + \gamma\|Dx\|_{\text{iso}}, \tag{4}$$

and second,

$$\min_x \|Kx-b\|_2^2 + \delta_S(x) + \gamma\|Dx\|_{\text{iso}}, \tag{5}$$

where $S = \{x \mid 0 \leq x \leq 1\}$, so that δ_S aims to restrict the range of allowed pixels. The iso norm is defined as

$$\|(x,y)\|_{\text{iso}} = \sum_{k=1}^{n} (x_k^2 + y_k^2)^{\frac{1}{2}}$$

2.1 Primal Douglas-Rachford splitting (Spingarn's method)

This first algorithm, which we will refer to by PDRS, was invented in the 1950s as a method to solve the heat equation [1]. Per (4), let

$$\begin{aligned} f(x) =& \delta_S(x) \\ g(x) =& \|Kx-b\|_1 + \gamma\|Dx\|_{\text{iso}} \end{aligned} \tag{6}$$

and consider the minimization problem

$$\min_{x,y} f(x) + g(y) + \delta_0\left(\begin{bmatrix} A & -I \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}\right) \tag{7}$$

with the optimality condition

$$0 \in \partial f(x) + \partial g(y) + \partial \delta_0\left(\begin{bmatrix} A & -I \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}\right) =: F(x,y) \tag{8}$$

Since this particular problem is too computationally expensive, the strategy is to split F into $F = A + B$. Define

$$\begin{aligned} A(x,y) &= \partial f(x) + \partial g(y) = \partial(\delta_S(x)) + \partial(\|Ky-b\|_1 + \gamma\|Dy\|_{\text{iso}}) \\ B(x,y) &= \partial \delta_0\left(\begin{bmatrix} A & -I \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}\right) \end{aligned} \tag{9}$$

Acknowledgments

Please include a list of each person (with student ID) in your group and write a paragraph **for each person** with exactly what they did. Note that the Acknowledgment section does not count towards the page limit.

Example Acknowledgments would look like:

Alexandre St-Aubin (Student ID: 261115785) We ball.

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References

[1] Jim Douglas and H. H. Rachford. On the numerical solution of heat conduction problems in two and three space variables. *Transactions of the American Mathematical Society*, 82 (2):421, Jul 1956. doi: https://doi.org/10.2307/1993056. URL `https://typeset.io/pdf/on-the-numerical-solution-of-heat-conduction-problems-in-two-4lbdiqc5i8.pdf`.

[2] Daniel O'Connor and Lieven Vandenberghe. Primal-dual decomposition by operator splitting and applications to image deblurring. *SIAM Journal on Imaging Sciences*, 7(3):1724–1754, 2014. doi: 10.1137/13094671X. URL `https://doi.org/10.1137/13094671X`.

[3] Neal Parikh. Proximal algorithms. *Foundations and Trends® in Optimization*, 1(3):127–239, 2014. doi: https://doi.org/10.1561/2400000003.

A Appendix

Optionally include extra information (complete proofs, additional experiments and plots) in the appendix. This section will often be part of the supplemental material.